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Faculty of Computer Science and Mathematics
Artificial Intelligence for Infrastructure and Urban Development

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Extraction of infrastructure features at intersections from street-level imagery

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Business Understanding: Motivation

Use Cases

- **Urban planning and safety analysis**: Identifying bottlenecks, missing pedestrian facilities, or unsafe intersections.
- **Autonomous driving**: Providing machine-readable infrastructure representations for perception and navigation systems.
- **Digital infrastructure mapping**: Deriving up-to-date maps of lanes, sidewalks, bike paths, and crossings.

Goals

- data-driven pipeline for semantic segmentation of street-level imagery
- extraction of traffic-related infrastructure features

Data Understanding: Cityscapes Dataset



Figure 1: Cityscapes image location

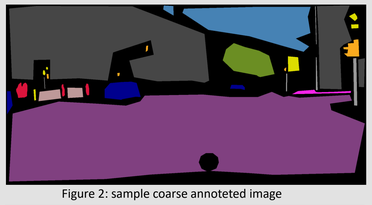


Figure 2: sample coarse annoteted image

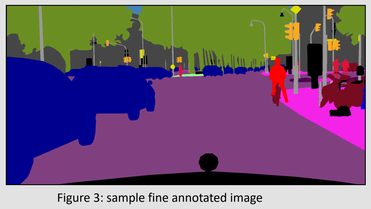


Figure 3: sample fine annotated image

Cityscapes Dataset

- Street level imagery
- 5,000 fine annotated images
- 20,000 coarse annotated images
- GPS-metadata
- 30 classes in total:
  - flat: road, sidewalk, parking, ...
  - human: person, rider (a person on a bycicle, motorcycle, scooter, ...)
  - vehicle: car, truck, bus, motorcycle, bycicle, ...
  - object: pole, traffic sign and traffic light, ...

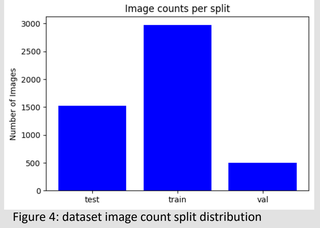


Figure 4: dataset image count split distribution

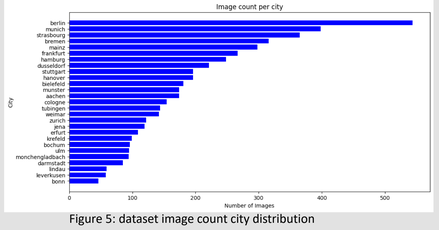


Figure 5: dataset image count city distribution

Data Preparation: Intersection Filtering

To isolate relevant images, multiple filters were applied:

- **Data Collection:**
  - Collecting intersection coordinates.
- **Filtering:**
  - Matching image GPS metadata (longitude, latitude) with the intersection data.
  - Filtering images only heading towards the intersection
- **Image count**
  - 25000 before -> after 3877 (coarse + fine)
- **Custom training classes**
  - From 19 to 20 classes (2 removed, 3 added)

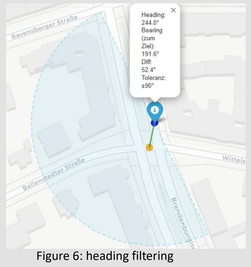


Figure 6: heading filtering

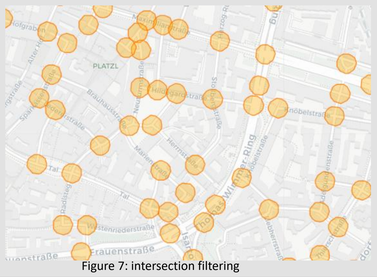
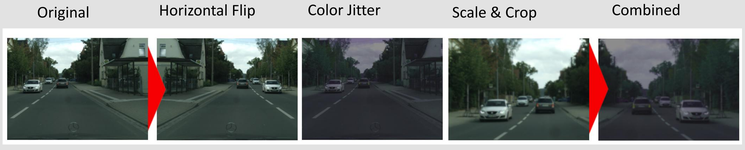


Figure 7: intersection filtering

Augmentation
To combat the small amount of available annotated images and to increase model generalization, augmentations were implemented.



Modeling

Models: U-Net, DeepLabv3, SegFormer
Custom loss function: Focal loss to combat class imbalance

2-Step-Training

- Pretraining with **coarse**
  - 15 epochs
  - 5 warmup epochs
  - Learning rate (0.001 to 0.00001)
- Training with **fine**
  - 150 epochs
  - 10 warmup epochs
  - Learning rate (0.001 to 0.00001)

Hyperparameters:

- batch_size 32
- AdamW optimizer with wheight_decay 0.0001

Evaluation Criterias:

- mIoU, pixel accuracy, loss

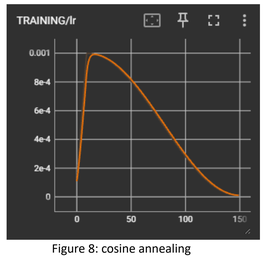


Figure 8: cosine annealing

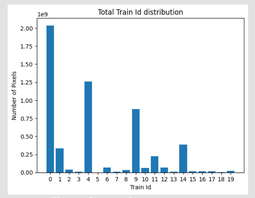


Figure 9: class imbalance

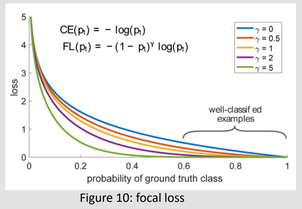


Figure 10: focal loss

Evaluation

Each model was trained on both the filtered and the complete Cityscapes dataset seperately.
The models were then evaluated on 4 self annotated mapilliary images of intersections in Regensburg.

Training on the intersections only dataset led to early overfitting and therefore decreased the models performance.



Figure 11: training and evaluation pipeline

The mIoU score is a more meaningful evaluation metric than pixel accuracy for image segmentation. Small pixel-level errors can lead to large differences in class overlap, especially for smaller classes.

DeepLabv3 trained on the complete Cityscapes dataset was the best model with the following scores:
train mIoU: 0.810, **val mIoU**: 0.634, **eval mIoU:** 0.388

	Model	Checkpoint	Train MIoU	Val (Cityscapes) MIoU	Eval (Regensburg) MIoU	Train Acc	Val (Cityscapes) Acc	Eval (Regensburg) Acc
0	deeplab	deeplab_cityscapes_3258105	0.694	0.509	0.318	0.960	0.929	0.841
1	segformer	segformer_3266966	0.558	0.467	0.333	0.933	0.912	0.855
2	unet	unet_3267195	0.347	0.328	0.247	0.882	0.874	0.821
3	deeplab	deeplab_cityscapes_giga_run_3271161	0.810	0.634	0.388	0.971	0.952	0.909
4	segformer	segformer_cityscapes_giga_run_3271909	0.688	0.599	0.389	0.955	0.943	0.903
5	unet	unet_cityscapes_giga_run_3272698	0.686	0.549	0.363	0.962	0.942	0.874

Figure 12: display of all relevant trained models

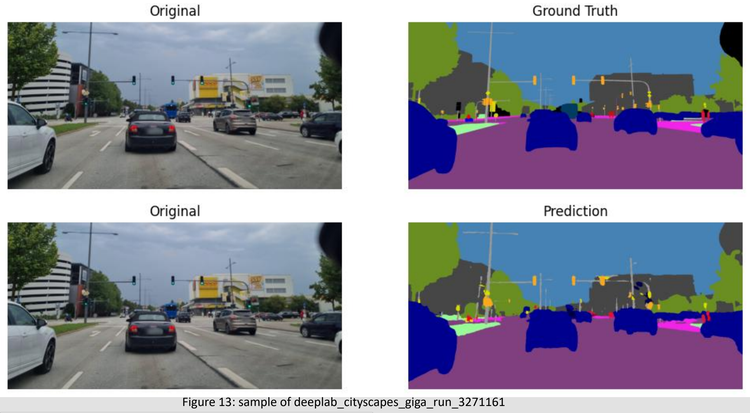


Figure 13: sample of deeplab_cityscapes_giga_run_3271161

Metrics

- Compute per-class instances from masks using CC, Watershed, or Peaks (fixed per class).
- Compare against manual ground truth to get absolute and % error per class.
- Determine a GT-weighted average % error across classes. (~**50%** for self annotated pictures)

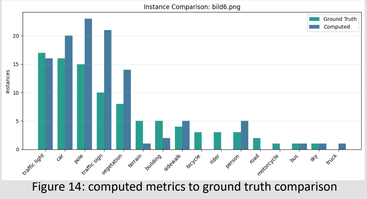


Figure 14: computed metrics to ground truth comparison